Bibek K C

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Summary

Full-Stack AI Engineer who enjoys the craft of shipping end-to-end AI products, from document understanding and RAG pipelines to agentic AI systems, desktop apps to web platforms, and air-gapped deployments to cloud-based enterprise solutions. At NeoALI, shipping on-device AI desktop applications used by 100+ users and building web-based AI solutions for office workflows, currently piloting with enterprise clients.

Feel free to chat with my AI assistant at kc-bibek.com.np.

Core Expertise

• AI Product Development • Agentic AI Systems • LLM Serving • Document Intelligence & RAG • Backend API Architecture • Enterprise & Air-Gapped Deployment • Desktop & Web Apps • Technical Leadership

Experience

AI Researcher & On-device Team Lead *NeoALI Co. Ltd, Korea*
Aug 2023 – Present

AI Products

- Technical lead of a 3-member on-device AI team, driving architecture decisions, deployment strategy, and cross-product engineering initiatives.
- Built and shipped SamuGen and SamuGen Plug-in, on-device AI desktop writing assistants used by 100+ users across air-gapped and B2C environments, supporting document-aware AI writing, Doc QnA, and HWP/HWPX/DOCX import/export, with a production-ready web version for on-premise deployment.
- Architected the backend of NeoClaw, an on-premise agentic proposal evaluation platform orchestrating scoring, summarization, synthesis, and ranking subagents, with QnA for querying AI scores.
- Designed and developed SlideGen, an on-premise LLM-powered slide generation and editing platform that creates presentations from uploaded documents with support for PPTX export.

Agentic Systems

- Built a unified agentic chat interface as the entrypoint across all on-premise AI products, with workflow orchestration that routes user intent to appropriate backends including proposal evaluation, RAG QnA, document translation, and other services.
- Developed a sandboxed Skills Server exposing pre-installed office skills (document creation and modification for DOCX, PPTX, XLSX, PDF) via APIs, eliminating runtime downloads to maintain security in enterprise web deployments.

AI Infrastructure

- Designed backend architecture using FastAPI, Express, Redis, SSE, authentication services, and API gateways, serving as shared infrastructure across multiple AI products.
- Operated shared LLM-serving infrastructure using vLLM, exposing OpenAI-compatible APIs for language models, embeddings, and reranking, serving as shared backend across 5 internal products. For on-device deployment, switched between IPEX-LLM Ollama, Llama-cpp, and OpenVINO.
- Contributed core components to ALIParse (in-house library), including layout analysis, reading-order ranking, document parsing, semantic retrieval, and RAG pipelines.

Enterprise Deployment

- Led deployment of AI desktop applications to 50+ air-gapped military notebooks, implementing offline license activation, model distribution, code obfuscation, and enterprise installation workflows.
- Built CI/CD pipelines, cross-platform packaging (Windows/macOS), monitoring dashboards, Slack-based alerting, load and stress testing, product download portals, and SMTP email workflows.

Research Intern *Robust Intelligence & Robotics Lab, KAIST*
Jul 2022 - Dec 2022 (6 months)

- For a single moving camera setup, developed a combined framework for real-time 6DoF Pose Estimation and Scene Graph Generation (SGG) in Robotic Manipulation Task using Cosypose framework for 6DoF pose and Neural-Motif framework for SGG and performed experiments on custom cup datasets. [Demo]
- For 2 fixed cameras setup, developed a probabilistic multi-view object-pose estimation framework that (i) associates multiple estimation results with scene graphs, (ii) combines pose distributions from singleview based estimators, and (iii) constructs a unified scene graph by predicting a unified pose distribution per object using MC dropout. More details in draft-paper.

Backend Developer Intern *Bisonai, Seoul*
Jan 2022 - Feb 2022 (2 months)

- Built data pipelines (DAGs) in Airflow to collect DeFi data from various sources like Uniswap, Coingecko and stored them in MongoDB Atlas.
- Implemented backend API server using FastAPI and Strawberry (Python GraphQL Library).

Data Visualization Project Intern *Interactive Computing Lab, KAIST*
Jul 2021 - Aug 2021 (2 months)

- Developed course materials for Data Visualization class, including tutorials for visualizing plots with Vegalite and D3.js in Observable notebook for beginners to learn Data Visualization techniques.

Technical Skills

Languages	Python, TypeScript, JavaScript
Frontend	React, Next.js, Vite, Electron
Backend APIs	FastAPI, Express.js, GraphQL, Celery, SSE, WebSockets, Authentication/Authorization
AI & ML	PyTorch, vLLM, Ollama, IPEX-LLM, Llama-cpp, OpenVINO/OVMS, HuggingFace
Infrastructure	Docker/Compose, Nginx, Linux, GitHub Actions, Airflow, AWS, Google Cloud
Databases	Redis, PostgreSQL, MongoDB, Qdrant
Tools	Git, Github, Gitlab, Uptime-kuma, Locust, SuperTokens, PyInstaller, Inno Setup

Education

Korea Advanced Institute of Science and Technology (KAIST) *Sep 2019 – Aug 2023*
B.S. in Electrical Engineering & Computer Science

British Model College (BMC) *Jul 2015 – Jun 2017*
*GCE A Levels, 4A*s in Chemistry, Biology, Physics, Mathematics*

Awards

Intel AI PC Innovation Challenge - First Place	NeoALI, January 2025
AI Grand Challenge, IITP - Fifth Place	NeoALI, 2023
KAIPlus+ Scholarship Award	KAIST, Apr 2023
Social Innovation Award	Tsinghua University, Jul 2021
Inclusive Leadership	Common Purpose, 2021
KAIST International Student Scholarship	KAIST, 2019-2023
Best student of the batch 2015/2017	The British College, 2017
Country Topper in AS Level Chemistry	University of Cambridge, Nov 2016